

Charlottetown Road Network Analysis Report

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1. Introduction

Charlottetown is the capital city of Prince Edward Island, Canada, and is characterized by a relatively compact and spatially constrained urban form. Its road network provides a suitable case for studying how urban structure influences accessibility, connectivity, and vulnerability in small-scale cities.

In this study, the urban road system is modeled as a spatial graph, where intersections are represented as nodes and road segments as edges. The network is extracted from OpenStreetMap using Python (OSMnx) and analyzed using NetworkX. A set of graph-theoretic metrics is computed, including degree distribution, shortest path length, centrality measures, spectral properties, and network resilience under node removal. Together, these measures provide a multi-scale representation of the structural and functional characteristics of the urban network.

This study examines whether global connectivity in a small urban road network is structurally concentrated within a limited set of critical intersections, and how this structural hierarchy influences both accessibility and network resilience.

Rather than testing a strict hypothesis, this study adopts an exploratory network science framework aimed at identifying structural hierarchy, functional bottlenecks, and resilience patterns in spatially embedded urban road systems.

2. Literature Review

Urban road networks are commonly modeled as graphs, where intersections are represented as nodes and roads as edges. Porta, Crucitti, and Latora (2006) analyze urban street systems using network theory and introduce primal and dual graph representations. Their results show that

centrality measures can reveal the structural importance of streets and influence accessibility. Building on this, the present study applies similar methods to a smaller urban system and focuses on practical and operational implications for urban planning and infrastructure management.

Crucitti, Latora, and Porta (2006) further examine different centrality measures, including degree, betweenness, and closeness, and demonstrate that each captures distinct structural properties of urban networks. Their findings suggest that centrality distributions can differentiate urban forms. This report uses these indicators to identify key intersections in Charlottetown and assess their functional roles.

Barthelemy (2011) provides a broader theoretical framework for spatial networks, highlighting the importance of geographic constraints in shaping network structure. This perspective is particularly relevant for road systems, where spatial layout affects connectivity and accessibility. The current study applies these concepts to a real-world urban network.

Derrible and Kennedy (2010) investigate the robustness of urban rail networks by analyzing their response to node removals. Their results show that network resilience depends on redundancy and connectivity. Similarly, this report evaluates resilience by removing high-centrality nodes to examine potential fragmentation.

Boeing (2017) introduces OSMnx, a tool for acquiring and analyzing street network data from OpenStreetMap. This tool enables reproducible and scalable network analysis. In this study, OSMnx is used to construct and analyze the Charlottetown road network.

Finally, Lämmer et al. (2006) show that traffic flow in urban road networks is often concentrated on a small subset of streets, highlighting hierarchical traffic organization. This supports the idea that certain nodes are more critical than others. The present study identifies such critical nodes using centrality measures and examines their implications for network performance and maintenance planning.

Overall, existing studies focus on large urban systems or theoretical network properties, while this study applies these methods to a smaller city and emphasizes operational implications such as infrastructure maintenance.

3. Data and Methodology

The urban road network is modeled as a spatial graph, where intersections are represented as nodes (V) and road segments are represented as edges (E). The network is extracted from OpenStreetMap using Python (OSMnx) to ensure reproducibility and geographic accuracy.

To capture both structural and spatial properties, the network is represented as a weighted undirected graph $G = (V, E)$, where edge weights correspond to the physical length of road segments. This allows the analysis to reflect realistic travel costs rather than purely topological connectivity.

The weighted adjacency matrix A is defined such that A_{ij} represents the road length between nodes i and j if an edge exists, and 0 otherwise. The corresponding degree matrix D is defined as a diagonal matrix where each diagonal element corresponds to the sum of incident edge weights. The graph Laplacian is then defined as $L = D - A$, which is used for spectral analysis of global connectivity.

Shortest path distances are computed using Dijkstra's algorithm on weighted edges, ensuring that all path-based metrics reflect actual spatial distances. All analyses are conducted on the largest connected component of the network to maintain consistency and avoid disconnected subgraph distortion.

A set of complementary network metrics is used to capture different structural scales. Degree centrality captures local connectivity, closeness centrality measures global accessibility, and betweenness centrality captures flow dependency and identifies structural bottlenecks. Spectral properties derived from the Laplacian matrix are used to assess global connectivity and segmentation. Network resilience is evaluated through node removal experiments, where the impact on the size of the largest connected component is measured.

All centrality measures are computed using normalized implementations provided by NetworkX to ensure comparability across networks of different sizes. The analysis is implemented using Python libraries including OSMnx, NetworkX, NumPy, SciPy, and Matplotlib. This framework enables a multi-scale structural analysis of small urban road networks, capturing how spatial constraints shape connectivity, accessibility, and network robustness.

4. Degree Distribution



Figure 1: Original road network of Charlottetown

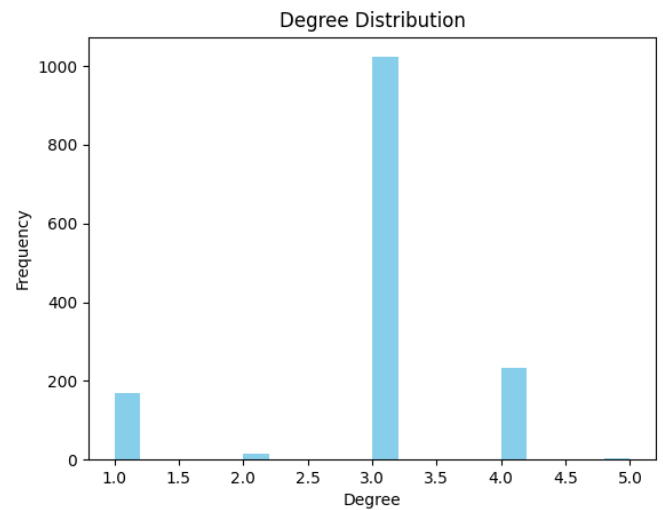


Figure 2: Degree distribution histogram

The degree of a node represents the number of incident edges connected to an intersection, reflecting local connectivity within the road network. Nodes with higher degree correspond to more complex junctions. Results show that the average degree is 2.918 and the maximum degree is 5.

Most intersections in Charlottetown have two or three connecting roads. This indicates a relatively regular and grid-like structure typical of small cities. Such a structure suggests balanced connectivity, where traffic can be distributed without relying on overly complex intersections.

The intersections with the highest degree are located along major roads such as St Peters Road, Mount Edward Road, and North River Road. These nodes serve as local hubs. Their disruption would affect access to multiple streets at once, making them important for maintaining local connectivity. This degree distribution reflects the structural constraints of planar spatial networks, where node connectivity is primarily governed by geometric embedding and road layout constraints rather than stochastic connectivity patterns observed in non-spatial random graphs.

5. Average Shortest Path Length

The average shortest path length measures the average travel distance between two intersections in the network. The computed average shortest path length (weighted by road length and calculated on the largest connected component) is 4142.242 meters.

This relatively small value reflects the compact geographic scale of Charlottetown and its well-connected road structure, where most locations can be reached within short travel distances, supporting efficient intra-urban movement.

The presence of multiple alternative short paths further reduces dependency on specific routes, improving overall accessibility and limiting travel detours across the network.

Overall, this pattern indicates that spatial efficiency is primarily driven by geographic compactness and local connectivity structure, a common characteristic of planar transportation networks where long-range shortcut edges are structurally constrained.

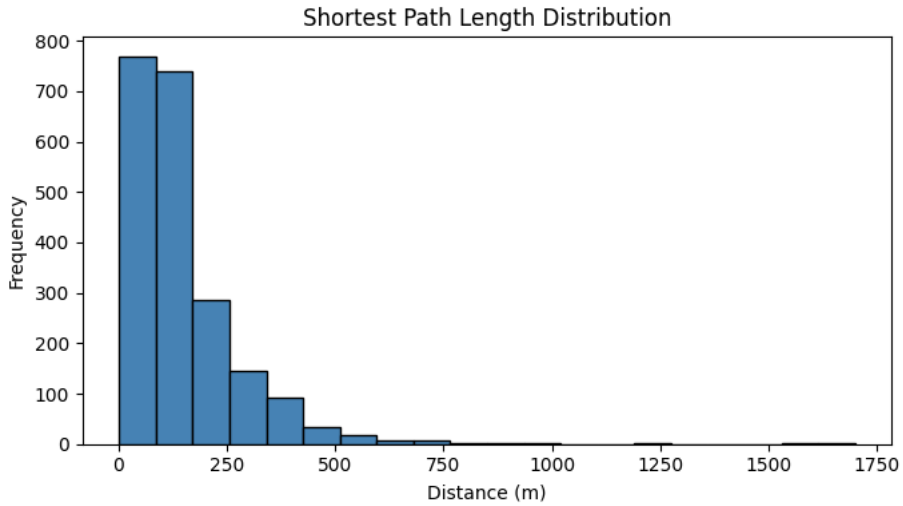


Figure 3: Shortest Path Length Distribution in Charlottetown (Quantifying urban compactness).

6. Betweenness Centrality

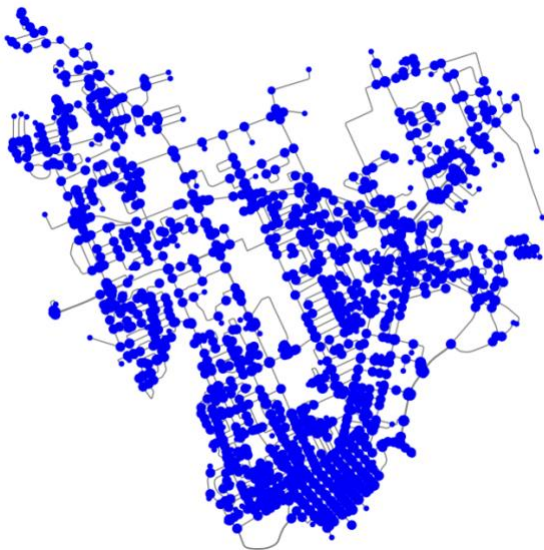


Figure 4: Node color by betweenness centrality

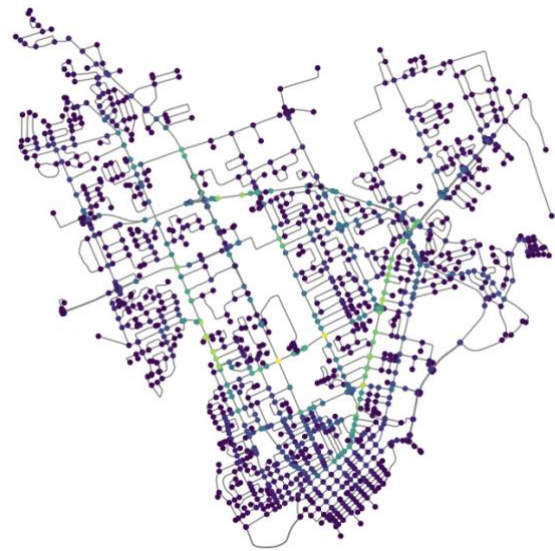


Figure 5: Top 5% betweenness nodes highlighted

Betweenness centrality captures the extent to which a node lies on shortest-path flows between all other node pairs, making it a measure of structural brokerage within the network. Nodes with high betweenness function as critical intermediaries that channel movement between otherwise weakly connected subregions.

The highest betweenness values are concentrated at key arterial intersections, including University Avenue and Belvedere Avenue, Belvedere Avenue and Mount Edward Road, and St Peters Road and Beasley Avenue. Additional high-centrality nodes are observed along North River Road, particularly at intersections with Warburton Drive and Waterview Heights.

These nodes form the backbone of shortest-path routing in the network, meaning that a large proportion of global flow is structurally dependent on them. Consequently, their removal would disproportionately increase path lengths and fragment local connectivity, highlighting their role as structural bottlenecks.

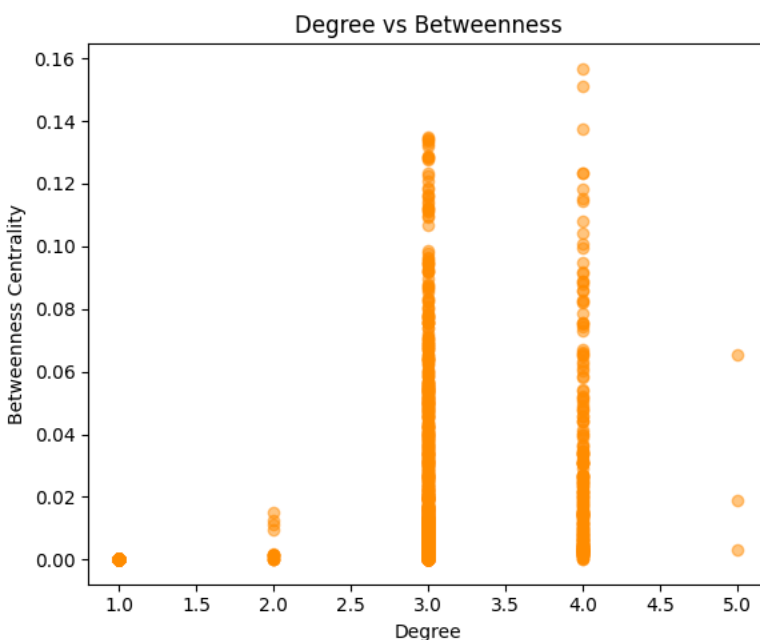


Figure 6: Scatter plot of Node Degree vs. Betweenness Centrality (Identifying structural bottlenecks).

7. Closeness Centrality

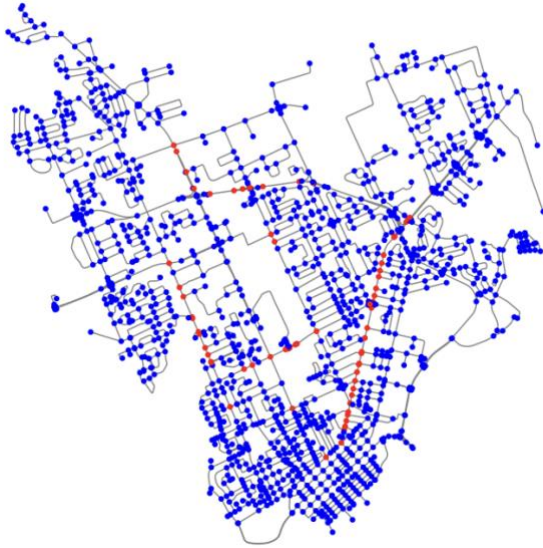


Figure 7: Node color by closeness centrality

Closeness centrality measures the average inverse shortest-path distance from a node to all other nodes, reflecting its overall accessibility within the network.

High closeness values are concentrated in the urban core, particularly around the downtown area, indicating that central locations are more efficiently integrated into the overall network structure. In contrast, peripheral areas exhibit lower accessibility due to increased path lengths to the rest of the network.

This spatial gradient reflects a hierarchical accessibility structure, where central nodes minimize global travel cost, while peripheral nodes are structurally constrained by their position within the planar network. This pattern highlights a core–periphery organization typical of spatially embedded urban road systems.

8. Spectral Properties

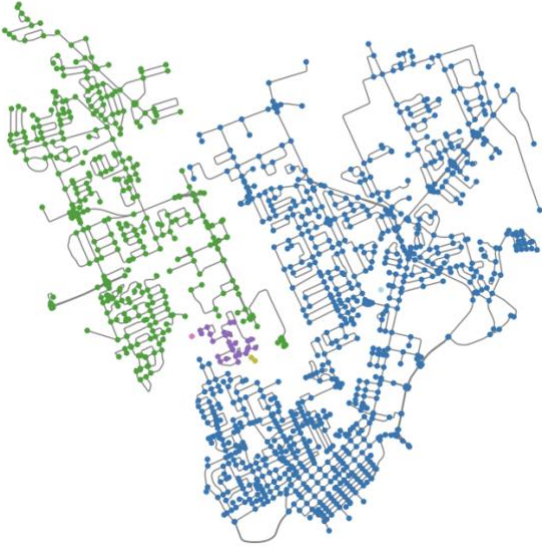


Figure 8: Node color by Fiedler vector

Spectral analysis of the road network is based on the graph Laplacian matrix, which captures global connectivity structure through its eigenvalue spectrum. The smallest non-zero eigenvalues are -6.57×10^{-16} , 0.001649, 0.001950, 0.003973, and 0.004641.

The second-smallest eigenvalue (algebraic connectivity, $\lambda_2 = 0.001649$) is relatively low, indicating weak global coupling between subregions of the network. This suggests that while the network is connected, its overall integration relies on a limited number of bridging structures, which is typical of spatially embedded planar graphs.

The Fiedler vector reveals a clear spatial partition of the network into functional regions, corresponding to the urban core, arterial corridors, and peripheral residential zones. This indicates a latent community-like structure induced by spatial constraints, where connectivity between regions is mediated by a small set of linking corridors.

9. Network Resilience

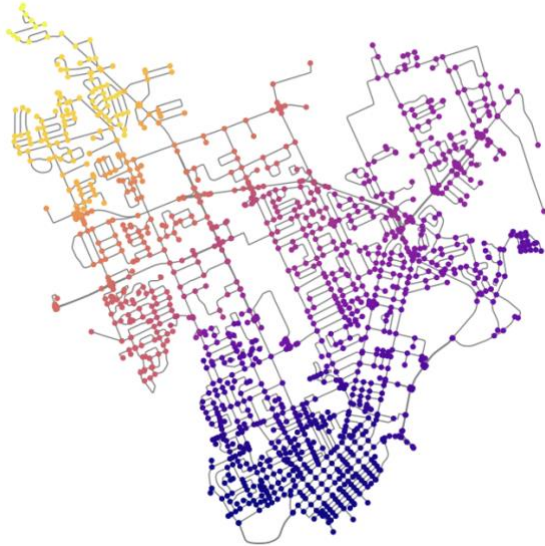


Figure 9: Network fragmentation after removing critical nodes

Network resilience is evaluated by simulating targeted node removal based on betweenness centrality and measuring the resulting fragmentation of the largest connected component. This is compared against a random failure baseline of equal size.

Results show that targeted removal of high-betweenness nodes leads to substantially greater fragmentation than random removal, confirming that global connectivity is disproportionately dependent on a small set of structural bottlenecks.

Although the absolute number of disconnected nodes remains limited, the affected nodes are concentrated in peripheral areas with low redundancy, where alternative routing options are scarce. In contrast, the core network remains largely intact due to higher path redundancy.

This asymmetry indicates a heterogeneous resilience structure: central regions exhibit high robustness due to dense connectivity, while peripheral regions are structurally fragile.

The contrast between targeted and random failure highlights the presence of a “critical backbone” in the network, primarily formed by high-betweenness arterial intersections that maintain global reachability.

10. Implications for Snow Removal Planning

The network-based indicators provide a quantitative framework for optimizing snow removal operations in Charlottetown. Rather than treating the city as a uniform spatial surface, centrality measures identify a hierarchical structure in which certain intersections play disproportionate roles in maintaining accessibility.

Betweenness centrality highlights critical intersections that function as connectivity bottlenecks, where clearance would restore access across multiple network segments. Closeness centrality identifies areas of high accessibility that enable efficient city-wide traversal, while spectral and resilience analyses reveal structurally dependent regions that are more sensitive to disruption.

Taken together, these measures suggest that operational prioritization should focus on maintaining connectivity along key arterial corridors, which form the structural backbone of the network. This framework can also be extended to routing and resource allocation models under uncertainty, improving operational efficiency and reducing response time during extreme weather events.

11. Conclusion

Metric	Value	Interpretation
Nodes	1444	Total intersections in Charlottetown road network
Edges	3762	Total road segments
Largest Connected Component	1444 nodes	Fully connected network
Average Degree	2.918	Sparse but well-connected urban structure
Maximum Degree	5	Highest intersection complexity
Average Shortest Path Length	4142.242 m	Short travel distances due to compact city layout
Algebraic Connectivity (λ_2)	0.001649	Low but stable global connectivity

Betweenness Removal Impact	6 nodes disconnected (n=72)	High redundancy in core network
Top Betweenness Node	0.1568	Key connector at University Ave – Belvedere Ave
Top Closeness Value	0.0003318	Most accessible central node
Network Type	Spatial road graph	Urban street network (OSMnx-based)

This study analyzed the structural properties of the Charlottetown road network using a combination of graph-theoretic and spectral methods. The results indicate that the network exhibits a sparse but highly structured topology, characterized by short average path lengths and low but sufficient global connectivity.

Centrality-based measures identify a small subset of intersections that play a disproportionate role in maintaining network cohesion. Spectral analysis further reveals a weakly integrated structure, where global connectivity depends on a limited number of bridging links between spatial subregions.

Resilience analysis confirms that the network remains connected under random failures but is significantly more sensitive to targeted removal of high-betweenness nodes, indicating heterogeneous robustness across different spatial zones.

Overall, the results highlight a fundamental trade-off in small urban road networks between spatial efficiency and structural vulnerability. While compactness ensures low travel costs and high accessibility, the reliance on a small number of high-betweenness intersections introduces localized fragility under targeted disruptions. This duality suggests that urban road systems are best understood as hierarchically organized spatial networks rather than uniformly connected grids, with important implications for infrastructure planning and resilience management.

12. Snow Removal Operations in Charlottetown

Snow removal in Charlottetown is governed by the **Winter Maintenance Operations Policy (City of Charlottetown, 2024)**. Managed by the Public Works Department, this operational framework is designed to balance public safety with budgetary

constraints, prioritizing high-volume infrastructure to maintain essential socio-economic activity (City of Charlottetown, 2023).

12.1 Road Priority Hierarchy

The city's snow clearing strategy follows a **tiered service-level agreement (SLA)** based on road classification defined by the provincial transportation standards (Government of PEI, 2022):

- **Priority 1: Arterial Roads:** Defined as "high-volume through-streets" (e.g., University Avenue, St Peters Road), these must be cleared to bare pavement within a specific response time window following a storm event (City of Charlottetown, 2024).
- **Priority 2: Collector Roads:** These connect residential feeders to the arterial spine and are cleared once Priority 1 routes reach a safe functional state.
- **Priority 3: Local Streets:** These areas are serviced last, with the city's policy allowing for "snow pack" conditions in specific sub-zero temperatures (City of Charlottetown, 2024).

This hierarchy reflects a functional dependency structure where a subset of arterial roads maintains global connectivity across the city.

12.2 Operational Workflow

The snow removal process typically follows a structured sequence:

1. Weather monitoring and storm classification
2. Activation of winter maintenance operations
3. Enforcement of parking restrictions in designated zones
4. Plowing of arterial roads (Priority 1 network)
5. Treatment of collector roads (Priority 2 network)
6. Residential street clearance (Priority 3 network)
7. Snow hauling operations in high-density or downtown areas

Snow hauling is particularly important in central Charlottetown due to limited storage space, requiring additional logistics beyond standard plowing operations.

12.3 Spatial Inequality in Service Response

The operational structure inherently creates spatial differences in service timing. Central arterial corridors are maintained at higher service levels and faster response times, while peripheral residential zones experience delayed clearance.

This creates a functional hierarchy where accessibility is not uniformly distributed but instead depends on the classification of the road within the operational network.

13. Comparison with Network Science Results

The empirical snow removal system can be directly compared with the structural properties of the Charlottetown road network derived from graph-based analysis.

13.1 Correspondence with Betweenness Centrality

The empirical prioritization of arterial corridors in Charlottetown demonstrates a strong correspondence with network flow theory (Porta et al., 2006), where betweenness centrality identifies nodes that mediate shortest-path flows and form the structural backbone of urban movement.

According to the City of Charlottetown Winter Maintenance Operations Policy (2024), key arterial roads such as University Avenue, Belvedere Avenue, and St Peters Road are classified as Priority 1 routes. These corridors are cleared first during winter storm events to maintain city-wide accessibility, emergency response capacity, and commuter flow.

From a network science perspective, these same intersections correspond to the highest betweenness centrality values in the constructed road network. This indicates that they function as structural bottlenecks through which a large proportion of shortest-path routing is concentrated.

This alignment suggests that municipal snow removal strategy implicitly targets the network's critical connectivity backbone. Disruption of these nodes would significantly increase global shortest-path distances and reduce overall network accessibility, confirming their role as essential infrastructure components in maintaining urban connectivity.

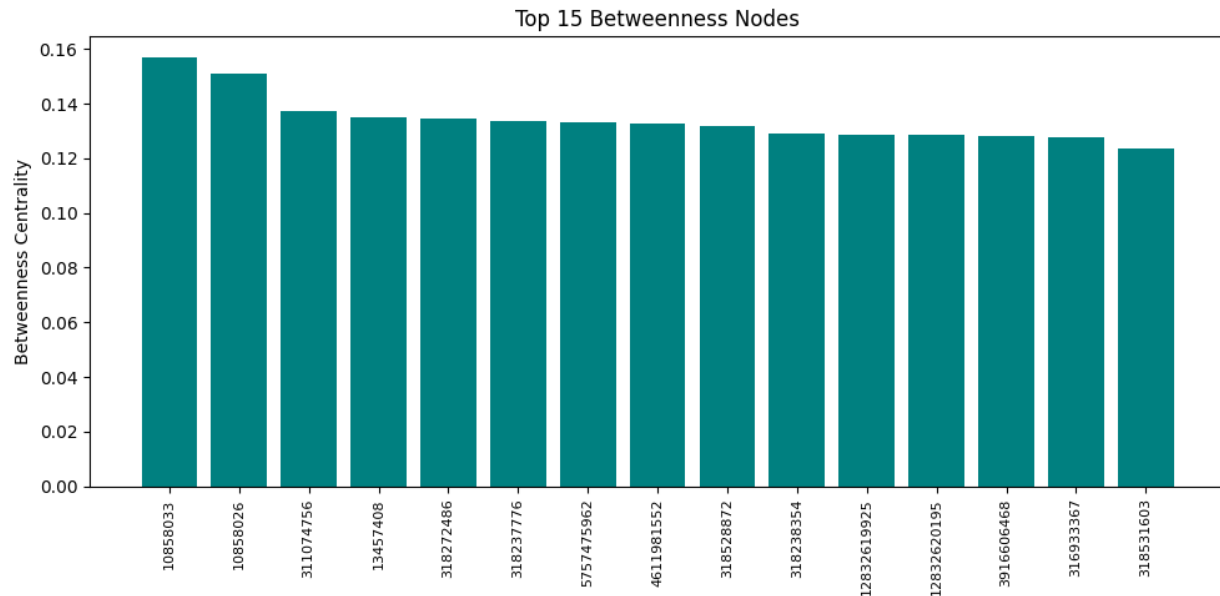


Figure 10: Ranking of Top 15 Intersections by Betweenness Centrality (Validation of Priority 1 routes).

13.2 Closeness Centrality and Operational Accessibility

Closeness centrality is highest in the urban core, particularly around downtown Charlottetown. This indicates that central nodes minimize average travel distance to all other nodes. This aligns with the city’s operational strategy, where downtown and arterial zones receive faster clearance due to their high accessibility importance. Peripheral areas, which show lower closeness values, correspond to lower-priority residential snow removal zones.

13.3 Spectral Structure and Regional Partitioning

Spectral analysis based on the Laplacian matrix reveals a weakly coupled network structure with clear spatial partitioning into:

- Urban core
- Arterial corridors
- Peripheral residential zones

The Fiedler vector indicates that connectivity between these regions is mediated by a limited number of bridging corridors. This matches operational reality, where a small set of arterial roads connects otherwise separated residential clusters.

13.4 Network Resilience and Snow Removal Dependency

The fragmentation observed in the resilience analysis reveals that the removal of top-centrality nodes leads to the isolation of peripheral clusters. This structural vulnerability finds a practical counter-measure in the Emergency Access Route protocols (City of Charlottetown, 2023), which prioritize maintaining a "functional core" during extreme weather. However, a critical gap exists where the graph-theoretic findings intersect with current policy. While the city aims to prevent total isolation, the existing Priority 3 classification for peripheral zones may underestimate the risk of "topological disconnection" as defined by Derrible and Kennedy (2010). This research suggests that even if these areas are not physically blocked, their reliance on a small set of high-betweenness bridges creates a "resilience deficit" that warrants a re-evaluation of service level agreements for secondary residential links.

14. Discussion: Alignment Between Policy and Network Structure

The comparison between empirical snow removal operations and graph-theoretic analysis reveals a strong structural alignment:

1. Arterial prioritization corresponds to high-betweenness nodes
2. Downtown priority corresponds to high-closeness regions
3. Peripheral delay corresponds to low-centrality regions
4. Connectivity backbone corresponds to spectral structure

This suggests that the city's operational system is not arbitrary but is implicitly optimized around the underlying topology of the road network.

However, the network analysis also reveals structural insights not explicitly considered in operational planning:

- Some peripheral regions show high vulnerability despite low traffic importance
- Certain bridging nodes carry disproportionate flow load
- Redundancy is unevenly distributed across spatial zones

These findings suggest potential improvements in resilience-based planning.

15. Conclusion (Section-specific)

The snow removal system in Charlottetown follows a hierarchical operational structure that closely aligns with the underlying topology of the urban road network. Graph-based analysis reveals that this hierarchy corresponds to measurable structural properties including betweenness centrality, closeness centrality, and spectral partitioning.

While the current system is effective in maintaining global connectivity, it relies heavily on a limited set of arterial corridors, creating a structurally efficient but spatially heterogeneous system. This demonstrates how network science methods can provide quantitative validation and extension of municipal infrastructure strategies.

Reference:

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